QP CODE-**25901261**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCFSQ200 - SENSORYSCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutesMaximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Label the test that measures force, stress, and strain.
   c) Fundamental
2. Indicate why sample containers should be odour-free in sensory tests.
   b) Flavour contamination
3. Recognize the principle that ensures comparability between samples.
   a) Random serving order
4. Select the characteristic flavor note of ripe fruits:
   b) Sweet and aromatic
5. Recall what the panelist is asked to do in a descriptive test.
   d) Describe intensity of each attribute
6. Select the statement that best distinguishes sensory evaluation from instrumental analysis.
   b) Sensory evaluation depends on human perception
7. Preference tests mainly measure
   c) Consumer choice
8. The sensor array in E-nose consists of
   c) Multiple sensor
9. State the most suitable temperature range for sensory evaluation rooms.
   d) 18–22 °C
10. Name the component that stores and displays analyzed data
    b) Computer system

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

11. Outline the steps involved in the mechanism of taste perception.

Taste perception occurs through the following steps:

1. Dissolution of food substances – Food chemicals dissolve in saliva in the mouth.
2. Contact with taste receptors – Dissolved substances interact with taste receptor cells located in taste buds on the tongue.
3. Stimulation of taste cells – Chemical substances stimulate receptor cells corresponding to basic tastes such as sweet, sour, salty, bitter, and umami.
4. Transmission of nerve signals – The stimulated receptors send signals through sensory nerves to the brain.
5. Interpretation by the brain – The brain processes these signals and identifies the taste sensation.

12. Define orthonasal smelling. Describe its role in odour perception.

Orthonasal smelling is the perception of odours when volatile compounds enter the nose through the nostrils during inhalation.

Role in odour perception:

- It helps detect the aroma of food before it is eaten.
- It plays an important role in identifying food quality and freshness.
- It contributes to flavour perception along with taste.
- It influences appetite and food acceptance.
- It helps in detecting spoilage or undesirable odours in food.

13. Define odor and flavour. Explain their importance in food acceptance and consumer preference.

Odor: Odor is the smell produced by volatile compounds that are detected by the olfactory receptors in the nose.

Flavour:Flavour is the combined sensation of taste, smell, and mouthfeel experienced while eating food.

Importance in food acceptance and consumer preference:

- Odor and flavour strongly influence the attractiveness of food.
- Pleasant aroma increases appetite and enjoyment.
- They help consumers identify different foods and ingredients.

- Unpleasant odour or flavour can lead to rejection of food products.
- They play a key role in determining consumer preference and product success in the market.

14. Discuss mean and median with suitable examples.

Mean:
Mean is the average value of a set of numbers. It is calculated by dividing the sum of all observations by the total number of observations.

Formula:
Mean = (Sum of observations) / (Number of observations)

Example:
Scores given by panelists for sweetness: 6, 7, 8, 9, 10
Mean = (6 + 7 + 8 + 9 + 10) / 5 = 40 / 5 = 8

Median:
Median is the middle value in a set of observations arranged in ascending or descending order.

Example:
Scores: 3, 5, 7, 9, 11
The middle value is 7, so the median is 7.

Mean represents the average value, while median represents the central value of the data.

15. Describe factors influencing visual perception of food.

Visual perception of food refers to how the appearance of food influences sensory judgment.

Factors influencing visual perception include:

- Color: Bright and natural colors make food more attractive.
- Shape and size: Uniform shape and appropriate size improve product appeal.
- Surface appearance: Smoothness, glossiness, or texture affect perception.
- Clarity and transparency: Important in beverages and liquids.
- Lighting conditions: Proper lighting helps accurate observation of food appearance.

These factors influence the first impression of food and affect consumer acceptance.

16. Identify the components included in a sensory evaluation scorecard.

A sensory evaluation scorecard is used to record panelists' observations during sensory testing.

Components include:

- Sample identification number to identify each product sample.
- Sensory attributes such as appearance, colour, aroma, flavour, texture, and overall acceptability.
- Rating scale or score scale (for example 1–9 hedonic scale).
- Panelist details such as name or panelist number.
- Comments section for additional observations or remarks.

Part C

Essay

Answer any **2** questions

Each question carries 10 marks

17. List and describe the five basic tastes.

The five basic tastes are sweet, sour, salty, bitter, and umami. Sweet taste is usually produced by sugars such as glucose and sucrose and is generally pleasant, indicating the presence of energy-rich carbohydrates. Sour taste is caused by acids such as citric acid or acetic acid and gives a sharp or tangy sensation, commonly found in foods like lemon and vinegar. Salty taste is mainly produced by mineral salts, especially sodium chloride, and is important for maintaining electrolyte balance and enhancing flavor in foods. Bitter taste is caused by substances such as caffeine, quinine, and certain plant compounds; it is often considered unpleasant and may act as a warning signal for toxic substances. Umami taste is a savory taste produced by compounds such as monosodium glutamate (MSG) and naturally occurring glutamates in foods like meat, mushrooms, and cheese, giving a rich and meaty flavor.

18. Identify the seven types of rating tests used in sensory evaluation and explain any four.

Rating tests are used in sensory evaluation to measure the intensity or quality of sensory attributes of food products. The seven types of rating tests commonly used are numerical scale rating test, graphic rating test, category scale test, hedonic rating test, ranking test, paired comparison test, and composite scoring test.

The numerical scale rating test uses numbers, usually from 1 to 9 or 1 to 10, to rate the intensity or quality of a sensory attribute such as sweetness or texture. In the graphic rating test, panelists mark their response on a line between two extremes, such as “very poor” and “excellent,” to indicate the intensity of a characteristic. The category scale test provides specific categories such as poor, fair, good, very good, and excellent, and panelists select the category that best represents the sample. The hedonic rating test measures the degree of liking or preference of a product, commonly using a 9-point hedonic scale ranging from “dislike extremely” to “like extremely.” These tests help researchers evaluate consumer acceptance and sensory quality of food products.

19. Explain the importance of texture in processed foods.

Texture is an important sensory attribute that refers to the physical feel and structure of food when it is handled or eaten. It includes characteristics such as hardness, softness, crispness, chewiness, and smoothness. In processed foods, texture plays a major role in determining consumer acceptance and overall product quality. A desirable texture enhances the eating experience and contributes to the perception of freshness and quality. For example, crispness is expected in chips, softness in bread, and creaminess in dairy products. Improper processing, storage, or formulation can negatively affect texture, leading to undesirable characteristics such as staleness or toughness. Therefore, maintaining the appropriate texture during processing and storage is essential for the success of processed food products.

20. Discuss the steps involved in recruiting sensory panel members for a laboratory.

Recruiting sensory panel members is an important step in conducting accurate sensory evaluation. The first step is identifying the need for panelists based on the type of sensory test and the number of participants required. The next step is advertising or inviting volunteers from students, staff, or the general public who are interested in participating in sensory testing. After this, screening tests are conducted to evaluate the sensory ability of the candidates, including their ability to detect basic tastes, odors, and differences between samples. Candidates who perform well in these tests are selected for the panel. The selected individuals then undergo training sessions to familiarize them with sensory attributes, evaluation procedures, and the use of scorecards or scales. During training, panelists also learn how to avoid bias and maintain consistency in evaluation. Finally, the trained panelists are organized into a sensory panel and are regularly monitored and evaluated to maintain their performance and reliability in sensory testing. Proper recruitment and training ensure accurate and reliable sensory evaluation results.